A study of Nuclear Gamma Spectroscopy using HPGe Detector

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A study of Nuclear Gamma Spectroscopy using HPGe Detector

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The methods of gamma spectroscopy were studied using a p-type coaxial HPGe detector. Measurement specifications of the detector, including the calibration transformation function, the resolution and absolute efficiency at specific energy values were obtained. Spectral analysis was performed on the processed count spectra. With the understanding of the detector and gamma spectroscopy built up, two soils samples taken from the bottom of a Norwegian lake before and after the Chernobyl incident were studied with the radioactive nuclei contributing to the observation suggested. Traces due to the increase in ^{125}Sb and ^{137}Cs concentration were found with high confidence in the soil samples.

***Index Terms*—HPGe detector, gamma spectroscopy, Chernobyl radionuclides.**

I. Introduction

The aim of the experiment was to identify the radioactive composition of a given sample, together with other radiation sources in the surroundings, from a measured gamma ray energy spectrum. This experiment was conducted in a normal laboratory setting. It was assumed that the sample was free from high energy radiation and would mainly undergo spontaneous decay. As a result, the main focus of electromagnetic radiation was the gamma ray emitted following beta decay [1].

High purity germanium detector contains germanium crystal as a semiconductor diode. With no impurity, the covalent and ionic characteristics of semiconductor allow the material to have two energy bands, the lower energy valence p-band and the higher energy conduction n-band. This leads to the presence of charge carrier, an electron-hole pair. This pair is formed when a intrinsic particle is collided by a photon with energy greater than the band gap, defined as the difference in energy between the two bands. When there is an electric field present at the charge carriers, the negatively charged free electron and the positively charged electron-accepting ion will move in opposite direction with a net drift velocity. In a closed circuit, the two charge carriers will move away from each other until they have reached the potential boundaries. To apply this property in gamma spectroscopy, a high voltage is applied between the crystal so the distance between the potential increases, leaving a larger volume of intrinsic layer in the p-i-n semiconductor. [2]

With the presence of high voltage, there is a source of significant systematic error known as leakage current [2]. Due to the weak conducting property being magnified by strong electric field, random fluctuation in the kinetic energy of atoms leading to excitation of electrons will result in noise being observed in the signal. Therefore, the semiconductor crystal has to be cooled to low temperature to minimise the probability of such event. Other practical additions to the detector system include a preamplifier and an amplifier, applied to the output signal from the detection crystal. The detailed descriptions can be referred to the amplifier web page provided by ORTEC, which is a major manufacturer of nuclear spectroscopy equipment [3].

There is an alternative crystal for gamma spectroscopy, with the use of sodium iodide. Germanium atom in the crystal has a much lower band gap at $0.65 \pm 0.01\ eV$ [4], compared with that of NaI, at around $5.9\ eV$ [5]. In an ideal scenario where no Compton shift occurs, the number of charge carrier pairs produced, N, is the quotient of band gap, e_b, and the photon energy, E, satisfying $N = E/e_b$, as a result of chain interaction between atoms through excitations and deexcitations. This leads to the presence of more charge carriers in germanium than in NaI, under the same incident gamma radiation. Therefore, less information in this discrete description about the incident gamma ray is lost in germanium detector. To quantitatively justify the choice of solid detection medium, the resolution of a typical HPGe detector about 1.33 Mev ^{60}Co is $1.68\ keV$ [6], and that of a typical NaI detector is $6.7\ keV$ [7].

The output from the detector is electrical impulse with its current proportional to the energy of detected gamma ray. To make the results more accessible, the current pulses were collected and represented by a histogram, counting the frequency of different magnitudes. The process will give an uncalibrated count histogram for further spectral analysis.

II. Methodology

A. Materials, Equipment and Tools

Four radioactive samples were studied through gamma spectroscopy. Two of them, ^{60}Co and ^{207}Bi were used to calibrate the detector. The details of these two samples for calibration are shown in Table I. The other two were the soil samples taken from the basis of a Norwegian lake before and after the Chernobyl incident. They were the subjects of study, such that the radioactive nuclei related to the nuclear disaster were aimed to be identified.

The HPGe detector used was manufactured by ORTEC, having a model number of GEM 40P4-83. It was a p-type Coaxial HPGe Detector that includes the addition of lithium(Li) atom as a trivalent impurity. This lead to an increase in both of the density of electron acceptor and the total density of charge carriers [2]. The electricity conductivity in a p-type semiconductor is higher than that in a perfect intrinsic or compensated material, resulting in a stronger signal output from the detector.

An analogue to digital converter was used to transform the current received from the detector into discrete signal pulses.

The ADC was included in a digital signal processing device manufactured by ORTEC with model number ORTEC DSPEC LF. All of the factors in the ADC configuration were projected on the raw data collected since the detailed information about the controlled setup, such as the reference potential and the number of channels, was not given. Therefore, attention was raised for any potential uncertainties aroused in this stage. The signal processing software, MAESTRO Multi-channel Analyzer Emulation Software, was also built in the same device, keeping count record of the transformed pulses and return a histogram for further analysis. Due to finite computing power, there is a probability of insufficient time in one computing cycle, leading to a null results. The sum of these time slots were recorded as dead time and a special implementation was applied by the manufacturer [8].

With the obtained signal current count histogram from the processing device, data analysis was conducted on the uncalibrated histogram as raw data. Python programming language version 3.8.12 was used with basic scientific computing library [9] installed. A special package called "uncertainties v3.1.4" was used to perform error propagation [10]. Basic error propagation methods for multi-variable functions [11], as taking the Euclidean norm, was implemented to Python environment in this package.

TABLE I: The measured activity of radioactive sources used in the experiment [12] and their unique gamma ray energy values taken from the reference source [13]

Sample	Activity (MBq)	Measurement Date	Energy (keV)
^{60}Co	0.001490955	6 Jan 2016	1173.2
			1332.5
^{207}Bi	0.0339	16 Mar 2018	570
			1063
			1770

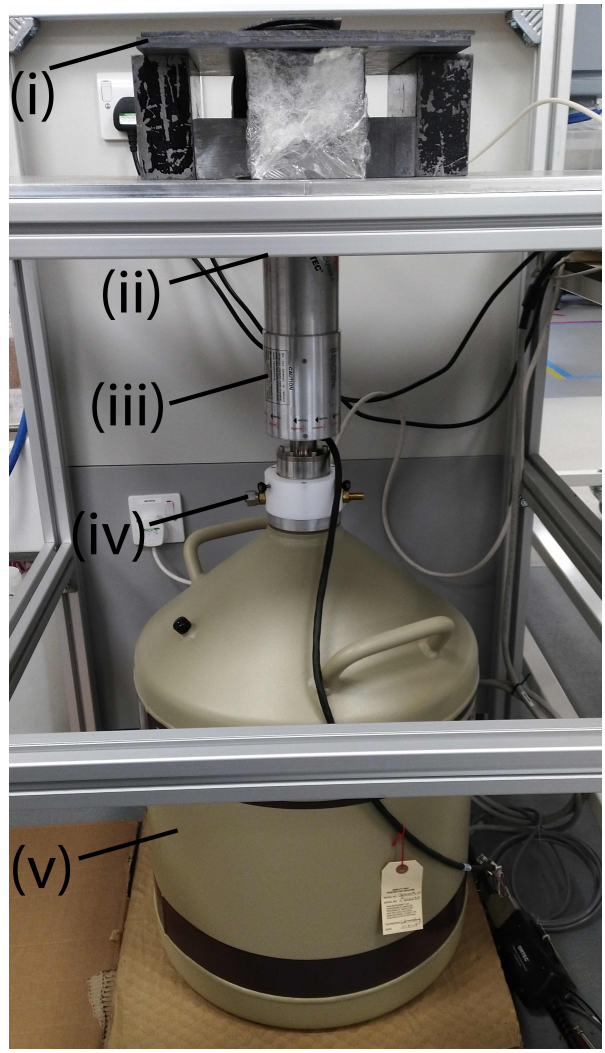

Fig. 1: Physical components of the HPGe detector with labels: (i) lead sheets, (ii) HPGe crystal, (iii) pre-amplifier, (iv) LN vent, (v) LN Dewar

B. Detailed Steps in the Experiment

The HPGe detector was first cooled it to low temperature. Liquid nitrogen at 77 K was pumped into the dewer 8 hours in advanced before the experiment started. A high voltage of 3000 V by setting was applied across the germanium crustal. The actual voltage was measured to be $(3130 \pm 20)\ V$, fluctuating in a small range. A minimum voltage of 500 V was included that the voltage across the crystal would never drop to zero during operation. A setup the equipment at this stage is presented in figure 1 and 2.

The radioactive samples for calibration, ^{60}Co and ^{207}Bi, were prepared for gamma spectroscopy. They were shielded with lead sheets to minimise background radiation. The measurements of sample for calibration were made in the time scale of about 20 minutes. Since the density of radioactive nuclei was high in the sample, the measurement time could be relative low as tens of minutes. On the other hand, the background radiation was measured with the absent of radioactive sample at the detection site, for a time scale of 1 day due to the much lower probability of incident photon. The measurement time was greatly extended to avoid an extreme case of Poisson statistics on small odd event. The same practice was repeated for the soil sample but in a greater extend, as it was expected to have similar count magnitude as its background radiation. The detailed measurement time values for the 4 samples, with live time and real time, are recorded in table II.

C. Data Analysis

Before any calculation were made, the uncertainties as the consequence of discrete measurement by counting bins in the histogram were identified as the major error in this experiment. The deviation from the true mean that in a single counting bin follows a Poisson distribution and is best estimated by equation (1). [14]

$$\sigma_N = \sqrt{N} \tag{1}$$

To calibrate the detector, the peaks in the histogram in logarithmic scale were first identified. Refer to equation (2),

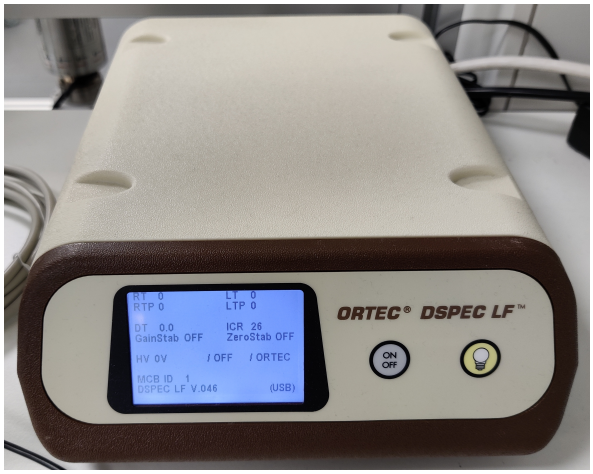

Fig. 2: User-interface of the signal processing device before high voltage is applied

TABLE II: Gamma spectroscopy measurement time values for the 4 radioactive samples and the corresponding radiation background conditions

Subject	Live time (s)	Real time (s)	Date
^{60}Co	884.62	983.84	14/11/2019
^{207}Bi	534.88	869.78	14/11/2019
background	98814.72	99302.14	13/11/2019
soil before Chernobyl	6.06×10^5	6.07×10^5	12/10/2021
soil after Chernobyl	3.34×10^5	3.35×10^5	12/10/2021
background	2.57×10^5	2.57×10^5	12/10/2021

a model as a Gaussian distribution on a linear straight line was used to apply linear regression fitting a small subset of the histogram near the observed peaks. The regression was performed using the "scipy.optimize.curve_fit" Python function [9]. These peaks were named as photo peaks and they corresponded to the sharp rise of count near the emitted gamma energy.

$$f(x; A, \mu, \sigma, m, c) = \frac{A}{\sigma\sqrt{2\pi}}\, exp(-\frac{1}{2}(\frac{x-\mu}{\sigma})^2) + mx + c \quad (2)$$

with variables defined as: $A \rightarrow$ Amplitude, $\mu \rightarrow$ Mean, $\sigma \rightarrow$ Standard Deviation, $m \rightarrow 1^{st}$ order linear parameter, $c \rightarrow 0^{th}$ order linear parameter.

After applying the fitting function to a photo peak in uncalibrated spectrum, the quantity μ of the Gaussian component, with its statistical error, would be obtained. These values were compared to the energy column in table I. With multiple fittings conducted to different peaks, these data points of μ were plotted in an uncalibrated channel against standardised energy axis. On this Energy-Channel plot, a linear polynomial, equation (3), was used as a fitting function to transform the channel onto a calibrated energy scale. Among different orders of linear function, the best fit function to these data points would be concluded as the calibration transformation.

$$P(x) = \sum_{i=0}^{\infty} q_i x^i \quad (3)$$

The overall resolution is defined in equation (4), where $FWHM$ refers to the full with of the gamma spectrum Gaussian peak at its half maximum and is approximated by $FWHM = 2.35\sigma$ [15]. The quantity of interest in the experiment was the effect on resolution due to electronics excluded from the detector. The resolution of the detector was provided and warranted by the manufacturer. It was considered constant so reducing the effect due to other factors would minimise experimental errors. The overall $FWHM$ was decomposed into $FWHM_d$ as the detector component and $FWHM_e$ as the electronics component, following the Euclidean norm and is presented in equation (5) [15]. Since the resolution was assumed constant, it was safe to apply the nonlinear response function proposed by Hakimabad [16], such that $FWHM_d$ was approximated to the first order of energy. Equation (6) was used to model the energy resolution function.

$$R_d = \frac{FWHM_d}{E_0} \quad (4)$$

$$FWHM^2_{overall} = FWHM^2_d + FWHM^2_e \quad (5)$$

$$FWHM_{overall} = \sqrt{CE_0 + FWHM^2_e} \quad (6)$$

A graph of $FWHM^2_{overall}$ against fitted energy values was plotted and fitted with a straight line using the same method as in calibration. The constant term C was referred to the slope and the terms contributing to the electronics resolution, $FWHM^2_{overall}$ was referred to the constant added.

The absolute efficiency was chosen to be the quantity of focus in the study of detection efficiency. It was defined by dividing the number of pulses recorded by the number of radiation quanta emitted by the source [15]. In addition, the peak efficiency was put be the interest as only interactions that deposit the full energy of the incident radiation contained most of the useful information for nuclear spectroscopy. In other words, the area under the fitted Gaussian curve was equivalent to the number of full energy interaction pulses recorded during the live time period. Refer to equation (2), since the Gaussian component was defined using the normalised form, the amplitude A had the same value as the area under the Gaussian curve. The number of radiation quanta emitted by the source could be calculated from the activity of the sample at the time of measurement. The activity was deduced using the inverse half life exponential decay model in equation (7) with reference to table I.

$$A = A_0(\frac{1}{2})^{t/h} \quad (7)$$

with variables defined as: $A \rightarrow$ Activity at time of measurement, $A_0 \rightarrow$ Activity at time of record, $t \rightarrow$ time since the record, $h \rightarrow$ the decay half life of nucleus.

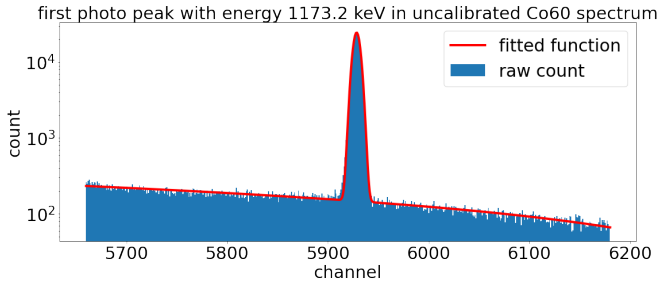

Fig. 3: expected typical fitting result with $\chi^2_{reduced} = 2.5$ and a live time of $884.62\ s$

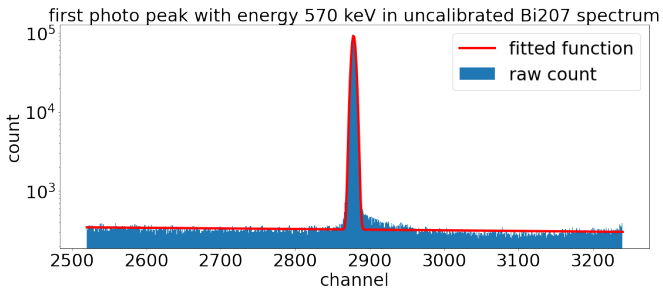

Fig. 4: unexpected fitting result with $\chi^2_{reduced} = 6.1$ and a live time of $534.88\ s$

III. Results

The calibration transformation was obtained as a linear function presented in equation (8).

$$f(x) = (0.1980 \pm 0.0001)x - (0.3 \pm 0.8) \tag{8}$$

An expected typical fitting to identify peak and an unexpected result are shown in figure 3 and 4. With the Gaussian mean values in the fitting to peaks in table III, first order polynomial was chosen to be the calibration function with reference to the Energy-Channel graph in figure 5.

Refer to equation (6), the linear factor, C, of energy, E, were found to be 0.0018 ± 0.0001. The FWHM due to electronics were calculated to have the value $(0.3 \pm 0.3)\ keV$. The proposed graph of $FWHM^2_{overall}$ against fitted energy values is shown in figure 8. The detector resolution at $1332.5\ keV$ was calculated to be $(1.56 \pm 0.07)\ keV$. By using linear interpolation, the $FWHM^2_{overall}$ at $112\ keV$ was predicted

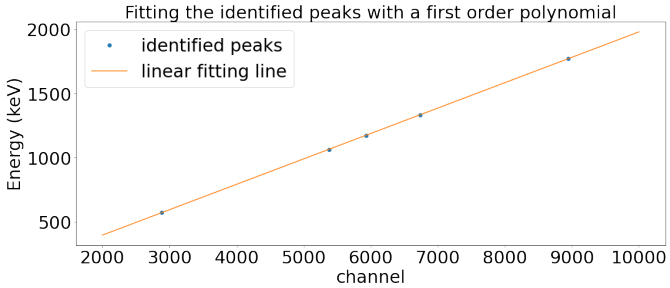

Fig. 5: Obtaining the calibration transform using a statistical approach, with $\chi^2_{reduced} = 0.3$

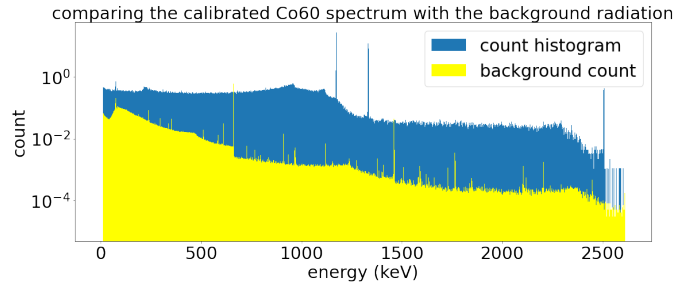

Fig. 6: Calibrated sample spectrum of ^{60}Co with the background, normalised by dividing count with live measurement time

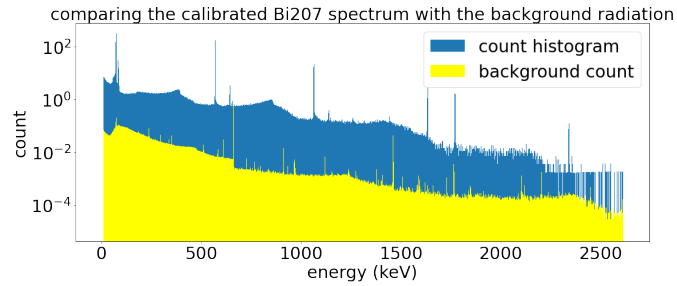

Fig. 7: Calibrated sample spectrum of ^{207}Bi with the background, normalised by dividing count with live measurement time

TABLE III: The results of applying linear regression on the identified peaks in the uncalibrated histogram

Identified Peak	Fitted Mean Parameter	$\chi^2_{reduced}$
^{60}Co Peak 1	5928.44 ± 0.01	2.5
^{60}Co Peak 2	6733.34 ± 0.01	2.2
^{207}Bi Peak 1	2878.35 ± 0.01	6.1
^{207}Bi Peak 2	5374.84 ± 0.01	4.0
^{207}Bi Peak 3	8945.70 ± 0.05	1.8

to be $(0.5 \pm 0.1)\ keV$. These results were reserved for further comparison with the manufacturer warranty.

An example calculation of the relative efficiency at $1332.5\ keV$ was performed. With the amplitudes of first and second peaks found to be 201700 ± 700 and 226800 ± 800, the relative efficiency was calculated to be $(47.1 \pm 0.2)\%$.

The soil samples' energy spectra were provided by a trusted source [17] using a different detector. With no information

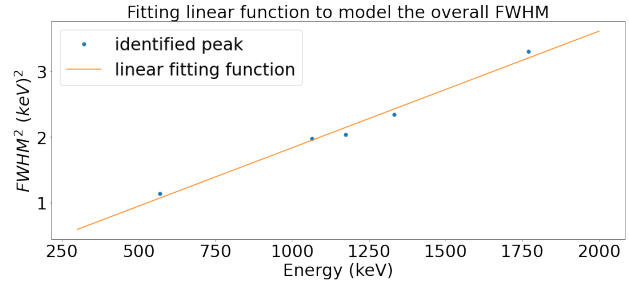

Fig. 8: linear fitting function applied to the plot of $FWHM^2_{overall}$ against fitted energy means, with $\chi^2_{reduced} = 0.01$

Identified Peak	Standard deviation	Amplitude
^{60}Co Peak 1	0.733 ± 0.002	226800 ± 800
^{60}Co Peak 2	0.777 ± 0.002	201700 ± 700
^{207}Bi Peak 1	0.581 ± 0.001	683000 ± 2000
^{207}Bi Peak 2	0.724 ± 0.002	298000 ± 1000
^{207}Bi Peak 3	0.898 ± 0.007	17600 ± 200

TABLE IV: The details of the resultant Gaussian peaks with the means transformed to calibrated values

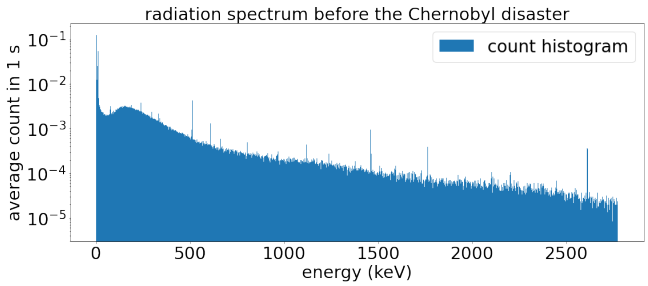

Fig. 9: gamma spectroscopy energy spectrum of the Norwegian soil sample before the Chernobyl disaster

regarding the efficiency at different energy values, it was not possible to calculate the radioactive composition by mass of the soil samples. Only hypothesis about the significant element contents could be introduced.

The normalised gamma spectra of them were shown in figure 9 and 10, with the background spectrum presented in figure 11. The radioactive nuclei content could be studied by subtracting the radiation spectrum by the background spectrum, shown in figure 12 and 13. An additional plot was made as figure 14 on the increase in soil radiation after the Chernobyl incident.

IV. Discussion

The fitting quality of the regression function was first assessed. From table III, the reduced chi square statistics were calculated to be in the range of 4 ± 2, indicating the fit was not ideal. A higher bin density in the fitting domain could be achieved by configuring a lower reference potential and a higher number of bins in the ADC. For the peaks in the ^{60}Co spectrum, the fitting function matched well with the

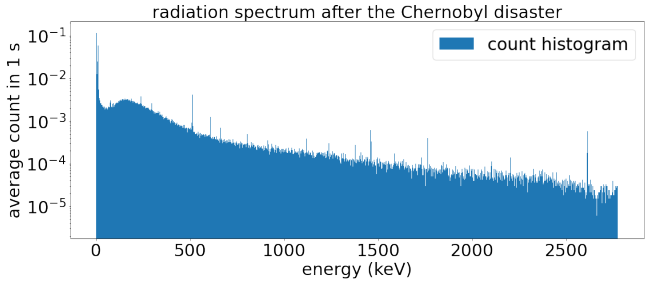

Fig. 10: gamma spectroscopy energy spectrum of the Norwegian soil sample after the Chernobyl disaster

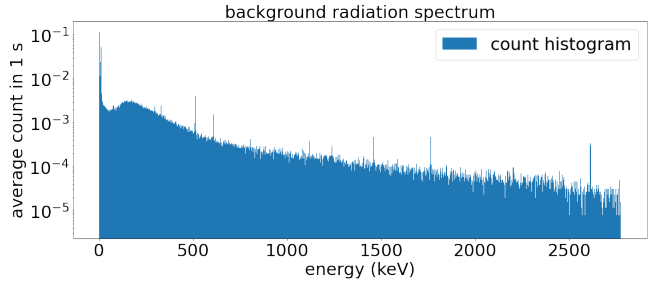

Fig. 11: gamma spectroscopy energy spectrum of the background radiation corresponding to the Norwegian soil sample

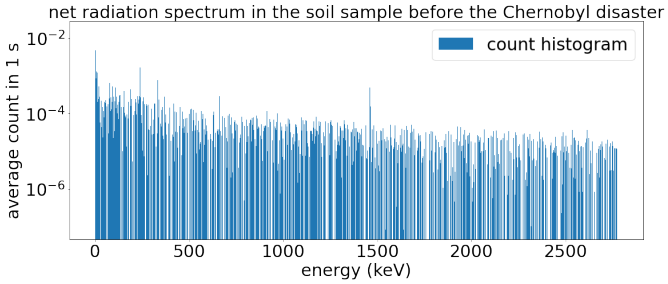

Fig. 12: energy spectrum of the difference between the background radiation and the emitted radiation from the Norwegian soil sample before the Chernobyl disaster

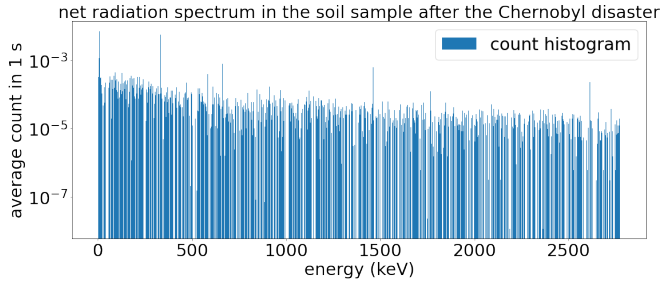

Fig. 13: energy spectrum of the difference between the background radiation and the emitted radiation from the Norwegian soil sample after the Chernobyl disaster

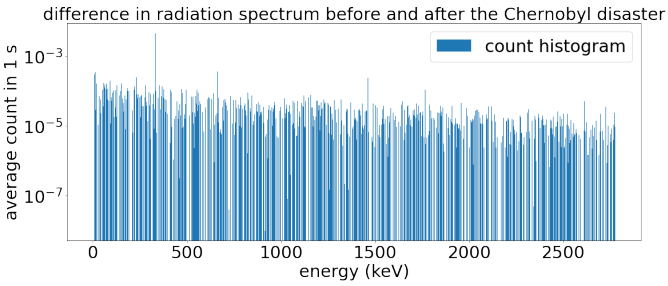

Fig. 14: energy spectrum of the difference between the emitted radiation from the Norwegian soil sample before and after the Chernobyl disaster

observation as in figure 3. For the peaks in the ^{207}Bi spectrum, the shape of the histogram peak was observed to have a tail on the higher energy side of the curve as in figure 4. This resulted in a much higher $\chi^2_{reduced}$ compared with the former. A new factor of the modelling how tilted the Gaussian curve could be introduced to improve the performance of fitting. With the fitted means obtained, the $\chi^2_{reduced}$ in the regression for calibration transformation was too low. Being at 0.3, it indicated over-fitting and could be resolved by introducing more radiation samples for calibration, such as ^{208}Tl and ^{88}Y.

In [18], the measured values of resolution and efficiency at $1330\ keV$ provided by the manufacturer were quoted to be $1.68\ keV$ and 41%. These quantities were calculated to be $(1.56 \pm 0.07)\ keV$ and $(47.1 \pm 0.2)\%$. The resolution at $122\ keV$ was provided as $680\ eV$ with the obtained value being $(500 \pm 100)\ eV$ in the experiment. A conclusion could be drawn that under the assumption that the condition of the detector was unchanged, the measurement errors in the connecting electronics was less significant in the conducted experiment, than in the measurement performed by the manufacturer. A higher performance at detecting high energy radiation was also observed. Therefore, it was concluded that the measurement produced by this HPGe detector could be trusted with relative high confidence. The measurement of the soil samples shall be repeated with the current detector with known performance specifications.

To comment on the provided data set of the soil samples, compare the the radiation spectrum before the incident with that afterwards, lower peaks were observed, showing the radioactivity was relatively low. Many of the nuclei with medium half-life, in a scale of 10^2 to 10^3 years, had decayed. This was expected to be observed in a typical soil sample as it has been exposed to the surrounding for a long time for equilibrium to be reached. One of the remaining nuclei was identified to be ^{40}K with gamma energy $1460\ keV$ and a long half-life of $1.3 \times 10^9\ years$ [13]. This trace of ^{40}K was also observed in the calibration sample spectra in figure 6 and 7. ^{40}K has a large abundance in the physical laboratory that even at a specially insulated Boulby Laboratory, its trace was still observed [19].

Compare figure 12 with figure 13, it was clear that the area under the curve is larger in the graph after the Chernobyl incident, indicating a higher radioactivity in the soil sample. This was further verified with a positive spectrum in figure 14, where the difference in activity before and after the incident was shown. In figure 14, three count peaks were observed with energy at around $332\ keV$, $662\ keV$ and $1460\ keV$. The nuclei causing these peaks were suggested to be ^{125}Sb, ^{137}Cs and ^{40}K. ^{125}Sb and ^{137}Cs are considered as parts of major Chernobyl fission products have a long decay half-life [20]. ^{40}K is a common decay end point that has a very low radioactivity. It can be affected by other geology factors such as the nuclear activity due to Earth's interior heat [21]. Therefore, it contributed less towards the radioactivity increase due to Chernobyl incident than ^{125}Sb and ^{137}Cs.

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